# NoctTwin architecture

How recorded nights become a bedtime risk list for clinicians

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Happiest Health Digital Twin Challenge 2026
Decision support prototype, not a medical device

1 · Data

### Dynamic stream

Continuous glucose every 15 minutes: 112,379 readings from 109 sessions.

Used only up to 22:00.

### Static stream

Labs, medications and demographics for each patient.

Tested in the ablation, where it added no measurable gain.

### ShanghaiT2DM

100 hospitalised patients. Public, anonymised data under CC BY 4.0.

Zhao et al., Scientific Data, 2023.

2 · Night builder

### Bedtime anchor at 22:00

Every feature uses only data recorded before the anchor.

Tonight: bedtime glucose, 1, 3 and 6-hour slopes, variability.

Earlier nights: prior low, nights since the last low, personal lowest value.

### Label and leakage guard

Persistent low: two or more consecutive readings below 70 mg/dL between 23:00 and 06:00.

Tests change the data after the anchor and check that features stay identical.

3 · Personal twin

### Synchronised forecast

The patient's own overnight profile plus a bedtime offset that fades through the night.

### Risk model

Calibrated logistic regression on CGM plus earlier nights. First nights use CGM only.

### Uncertainty and reasons

Split-conformal 80% and 90% ranges. Linear SHAP for each night's flag.

### What-if simulation

Two-compartment ODE changes. Effects are assumed and labelled as simulation.

4 · Clinician platform

### FastAPI service

Serves patients, nights, what-if scenarios and the model card from the generated result files.

### React web app

**Triage:** patients ranked by risk tier.

**Patient twin:** history, expected profile, range, reasons, what-if.

**Evidence:** results, limits, data provenance.

Runs from static files if the API is down.

### Clinician

Decision support, not a medical device.

## Validation, applied to every result

Patient-grouped 5-fold cross-validation, repeated 20 times

Paired patient-level bootstrap intervals

Calibration, Brier skill and conformal coverage on held-out patients

Tests: no leakage, canonical counts, headline-number guard